

Divisibility

Definition : Let $a, b \in \mathbb{Z}$. We say that b divides a , denoted $b|a$, if there exists an integer $k \in \mathbb{Z}$ such that $a = bk$. In this case, we call b a divisor of a , and a a multiple of b .

Notation : $b|a$ means ' b divides a '. Equivalently: $b \nmid a$ means ' b does not divide a '.

Examples : $3|12$ because $12 = 3 \cdot 4$.

$5 \nmid 12$ because there is no integer k with $12 = 5k$.

Question : Is 0 divisible by 0 ?

Properties of Divisibility

Theorem 1.1: Let $a, b, c \in \mathbb{Z}$. Then:

1. $a \mid a$ (reflexivity).
2. If $a \mid b$ and $b \mid c$, then $a \mid c$ (transitivity).
3. If $a \mid b$ and $a \mid c$, then $a \mid (bx + cy)$ for any $x, y \in \mathbb{Z}$ (linear combination).

Proof of (1) : $a = a.1$, so $a \mid a$.

Proof of (2) : If $a \mid b$, then $b = ak$. If $b \mid c$, then $c = bm = (ak)m = a(km)$, so $a \mid c$.

Proof of (3) : If $a \mid b$ and $a \mid c$, then $b = ak$ and $c = am$.

Thus $bx + cy = akx + amy = a(kx + my)$, so $a \mid (bx + cy)$.

Common Divisors

Definition : Let $a, b \in \mathbb{Z}$. An integer d is called a common divisor of a and b if $d|a$ and $d|b$.
The set of all common divisors of a and b is denoted $D(a,b) = \{d \in \mathbb{Z} : d|a \text{ and } d|b\}$.

Example: For $a = 12$ and $b = 18$:

Divisors of 12: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

Divisors of 18: $\pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18$

Common divisors $D(12,18) = \{\pm 1, \pm 2, \pm 3, \pm 6\}$

Note: Every pair of integers has at least ± 1 as common divisors.

Greatest Common Divisor

Definition: Let $a, b \in \mathbb{Z}$, not both zero. The greatest common divisor of a and b , denoted $\gcd(a, b)$, is the unique positive integer d such that:

- (1) $d \mid a$ and $d \mid b$ (d is a common divisor).
- (2) If $c \mid a$ and $c \mid b$, then $c \mid d$ (d is the greatest).

Equivalently: $\gcd(a, b) = \max\{d \in \mathbb{Z}^+ : d \mid a \text{ and } d \mid b\}$.

Example: $\gcd(12, 18) = 6$ because $6 \mid 12$ and $6 \mid 18$, and any common divisor of 12 and 18 divides 6.

Special case: $\gcd(a, 0) = |a|$ for any $a \neq 0$.

Codes

Divisibility

```
bool divides(long long b, long long a) {  
    return a % b == 0;  
}
```

Is there something
wrong with this
code?

GCD

```
long long gcd_naive(long long a, long long b) {  
    long long limit = min(a, b);  
    for (long long i = limit; i >= 1; i--) {  
        if (a % i == 0 && b % i == 0) {  
            return i;  
        }  
    }  
    return 1;  
}
```

$O(\min(a,b))$
Can we do better ?

The Euclidean Algorithm Property

Theorem (Euclidean Property) : Let $a, b \in \mathbb{Z}$ with $b \neq 0$. Then :

$$\gcd(a,b) = \gcd(b, a(\bmod b))$$

Proof : Let $r = a(\bmod b)$, so $a = bq + r$ where $0 \leq r < |b|$ and $q = \lfloor a/b \rfloor$.

We show $D(a,b) = D(b,r)$.

If $d \mid a$ and $d \mid b$, then $d \mid (a - bq) = r$, so $d \in D(b,r)$.

Conversely, if $d \mid b$ and $d \mid r$, then $d \mid (bq + r) = a$, so $d \in D(a,b)$.

Therefore $D(a,b) = D(b,r)$,

which implies $\gcd(a,b) = \gcd(b,r)$. ■

How it helps ?

- **Exponentially Shrinking Numbers** : Because we use the remainder at each step, the numbers being calculated shrink by at least half every two steps.
- **$O(\log N)$ Complexity** : In computer science, this rapid reduction means the algorithm runs in logarithmic time, making it exceptionally fast even for astronomically large numbers.
- **Lamé's Theorem (1844)** : A mathematical proof stating that the Euclidean algorithm will never take more steps than 5 times the number of digits in the smaller number.

Can you think of the worst case for this ?

The Euclidean Algorithm

- **Algorithm** : Euclidean Algorithm.
- **Input** : $a, b \in \mathbb{Z}$ with $a \geq b \geq 0$.
- **Output** : $\gcd(a, b)$.
- **Procedure** : while $b \neq 0$
do:
 $r \leftarrow a \bmod b$;
 $a \leftarrow b$;
 $b \leftarrow r$;
return a ;
- **Recursive form** : $\gcd(a, b) =$ (if $b = 0$) $\rightarrow a$ and (if $b \neq 0$) $\gcd(b, a \bmod b)$
- **Time Complexity** : $O(\log(\min(a, b)))$. This is because the Fibonacci sequence gives the worst case, and consecutive Fibonacci numbers decrease by the golden ratio.

Codes

GCD

```
long long gcd_recursive(long long a, long long b) {  
    if (b == 0) {  
        return a;  
    }  
    return gcd_recursive(b, a % b);  
}
```

$O(\min(\log(a), \log(b)))$

How can you find all the common divisors of A and B efficiently?

How can we find LCM of two numbers effectively ?

Least Common Multiple

Definition: Let $a, b \in \mathbb{Z}$, not both zero. The least common multiple of a and b , denoted $\text{lcm}(a,b)$, is the unique positive integer m such that:

- (1) $a \mid m$ and $b \mid m$ (m is a common multiple).
- (2) If $a \mid n$ and $b \mid n$, then $m \mid n$ (m is the least).

Equivalently: $\text{lcm}(a,b) = \min\{m \in \mathbb{Z}^+ : a \mid m \text{ and } b \mid m\}$

Theorem

For any $a, b \in \mathbb{Z}$ with $a, b \neq 0$:

$$\text{lcm}(a, b) \cdot \gcd(a, b) = |ab|$$

Proof: Let $d = \gcd(a,b)$, $a = da'$, $b = db'$ where $\gcd(a',b') = 1$. Then $\text{lcm}(a,b) = da'b'$ (since $a \mid da'b'$ and $b \mid da'b'$, and any common multiple must be divisible by $da'b'$). Thus $\text{lcm}(a,b) \cdot \gcd(a,b) = da'b' \cdot d = d^2a'b' = (da')(db') = |ab|$. ■

Codes

LCM

```
long long gcd(long long a, long long b) {  
    if (b == 0) return a;  
    return gcd(b, a % b);  
}
```

```
long long lcm(long long a, long long b) {  
    return (a / gcd(a, b)) * b;  
}
```

Time complexity ?

Note : We wrote $(a/\text{gcd}(a,b)) * b$ instead of $(a*b)/\text{gcd}(a,b)$. This is done to prevent integer overflow that might be cause when a and b are very large .

Bézout's Identity

Theorem (Bézout's Identity): Let $a, b \in \mathbb{Z}$. Then there exist integers $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$.

Proof (by strong induction on b):

Base case : If $b = 0$, then $\gcd(a, 0) = |a|$.

If $a > 0$, set $x = 1, y = 0$, so $a \cdot 1 + 0 \cdot 0 = a$.

Inductive step : Assume the theorem holds for all pairs with second element less than b .

For a, b with $b > 0$,

let $r = a \bmod b$ and $q = \lfloor a/b \rfloor$.

By induction, $\exists x_1, y_1 : bx_1 + ry_1 = \gcd(b, r) = \gcd(a, b)$.

Since $r = a - bq$: $bx_1 + (a - bq)y_1 = \gcd(a, b)$.

Rearranging: $ay_1 + b(x_1 - qy_1) = \gcd(a, b)$.

Set $x = y_1, y = x_1 - qy_1$. ■

Extended Euclidean Algorithm

- **Algorithm :** Extended Euclidean Algorithm
- **Input:** $a, b \in \mathbb{Z}$.
- **Output:** $\gcd(a,b)$ and integers x, y such that $ax + by = \gcd(a,b)$.
- **Procedure:** function $\text{extended_gcd}(a, b)$:
 - if $b = 0$:
 - return $(a, 1, 0)$
 - else:
 - $(d, x_1, y_1) \leftarrow \text{extended_gcd}(b, a \bmod b)$
 - $x \leftarrow y_1$
 - $y \leftarrow x_1 - \lfloor a/b \rfloor \cdot y_1$
 - return (d, x, y)
- **Correctness:** By Previous Theorem, coefficients x, y exist. The algorithm computes them via back-substitution during recursion unwinding.
- **Time Complexity:** $O(\log(\min(a,b)))$, same as standard Euclidean algorithm.

Extended Euclidean Algorithm Example

- **Problem :** Find integers x, y such that $35x + 15y = 5$

- **Formulas :** $x = y_1$ and $y = x_1 - y_1 \lfloor a/b \rfloor$

- **Recursive stack :**

$\text{ext_gcd}(35, 15) \rightarrow \text{ext_gcd}(15, 5)$

$\text{ext_gcd}(15, 5) \rightarrow \text{ext_gcd}(5, 0)$ (Base Case)

returns $d=5$, $x=1$, $y=0$;

- **Back Substitutions :**

for $a = 15, b = 5 \rightarrow x_1 = 1, y_1 = 0$. $x = 0$, $y = 1 - 0 * \lfloor 15/5 \rfloor = 1$.

for $a = 35, b = 15 \rightarrow x_1 = 0, y_1 = 1$. $x = 1$, $y = 0 - 1 * \lfloor 35/15 \rfloor = -2$.

Codes

Extended Euclidean Algorithm

```
long long extended_gcd(long long a, long long b, long long &x, long long &y) {  
    if (b == 0) {  
        x = 1;  
        y = 0;  
        return a;  
    }  
    long long x1, y1;  
    long long d = extended_gcd(b, a % b, x1, y1);  
    x = y1;  
    y = x1 - y1 * (a / b);  
    return d;  
}
```